

Mbarara Integrated Concrete Products Factory

Prompt Coverage, Module Specification & Integration Supplement

This supplement was added after review of the merged PDF to explicitly capture every prior prompt and requirement in one downloadable document. It is prepended to the full detailed investor report and recent module master report.

Currency: UGX only. Market and competitor values marked Estimated, Quotation Required, or Needs verification must be re-verified before investment decisions.

Prompt Coverage Matrix

The table below maps each user request to the corrected merged report and platform deliverables.

User request / request	Completed or corrected merged report	Platform deliverables
Professional 150-200 page investor-grade PDF report generator for the Mbarara Integrated Concrete Products Factory, UGX only.	Yes	Detailed investor report section appended after this supplement; full report is 194 pages and includes A4 portrait consultancy-report layout, table of contents, chapters, tables, charts, diagrams, image placeholders, and appendices.
Financial Simulator using React, Tailwind CSS, Recharts, local editable state, UGX inputs, product volumes/prices, cost/profit outputs, charts, CSV and PDF export.	Yes	Platform Financials tab, Product Costing tables, Financial Model chapter, ERP financial summary, and prompt coverage module specification.
Inventory Management System tracking cement, stone dust, sand, aggregates, diesel, water, pallets, moulds, spare parts; add/issue stock; low-stock alerts; supplier/truck/cost; valuation; monthly usage; reorder calculation.	Yes	Platform Inventory tab, PostgreSQL inventory and inventory_transactions tables, Raw Material Consumption chapter, Inventory module specification.
Quality Control Database tracking every production batch, material usage, mix ratio, curing, tests, pass/fail outputs, rejection rate, strength by product, defect trends, printable certificate.	Yes	Platform QC Lab tab, Production Batch register, quality_tests schema, Laboratory and Quality Control chapter, QC module specification.
Simple ERP system with Sales, Inventory, Production, Quality Control, Customers, Suppliers, Expenses, Reports; sales/production/report fields.	Yes	ERP tab, Database schema, ERP Integration Summary chapter, customer/supplier balances, daily sales, monthly profit, inventory usage, production efficiency.
Continue: next module integrated with existing ERP.	Yes	Dispatch & Delivery was added in the previous ERP iteration and documented here as part of ERP/dispatch integration history and operating workflow.
React + TypeScript + Tailwind CSS + Recharts, modular architecture, maintainability and scalability.	Yes	Platform architecture section, source files, components/common, layout, data, db, lib, modules structure.
PostgreSQL database structure for products, suppliers, customers, production batches, inventory, sales, expenses, quality tests; future ERP integration.	Yes	PostgreSQL tab and src/db/postgres-schema.sql; schema summary and relationship matrix in this supplement.
Inventory Management Module connected to Financial Simulator.	Yes	Shared AppState and computeErp/computeFinancials/computeInventory calculations; material prices and stock values feed financial reporting.
Quality Control and Laboratory Module integrated with production batches.	Yes	Batch IDs link production_batches to quality_tests; certificate uses batch/product/operator/approval data.
ERP module connected to sales, inventory, production, quality control, expenses.	Yes	ERP tab and PostgreSQL relationship matrix; reports combine sales, balances, stock usage, production efficiency, QC pass rate, expenses.
Competitor and Market Intelligence Database module with competitors, price intelligence, customer demand, gap scoring, SWOT, dashboard, focus areas, seed competitors, source tracking, CSV/PDF exports.	Yes	Market Intel tab, Market Analysis and Competitor Analysis chapters, Market Gap Analysis chapter, source-confidence notes, and export specification.
Reusable ReportPdfGenerator.tsx with jsPDF/pdfmake, page numbers, headings, tables, Recharts images, confidence labels, assumption labels, disclaimer, Download Master PDF button.	Yes	ReportPdfGenerator.tsx, Master PDF tab, saved downloadable PDF link, and this corrected merged PDF.

Module Specifications Included

Financial Simulator

Module	Specifications
Inputs	Cement price per 50 kg bag; stone dust, sand, aggregate per tonne; electricity per kWh; water per m3; diesel per litre; labour per month; transport per truck; working days; daily production; selling price per product.
Products	6-inch hollow blocks, 8-inch hollow blocks, solid blocks, 60 mm pavers, 80 mm pavers, kerbstones, drainage channels, culverts; broader portfolio also includes 4-inch blocks, grass/permeable pavers, cobblestone pavers, ready-mix concrete.
Outputs	Cost per product, monthly revenue, material cost, utility cost, labour cost, gross profit, net profit, break-even point, ROI estimate, and whether UGX 20,000,000/month is reached.
Charts	Revenue by product, profit by product, cost breakdown, monthly profit projection.
Exports	CSV export and PDF summary export in the market/financial reporting flow; master PDF consolidates final report.
Integration	Inventory prices and ERP sales/expenses share the same UGX model and calculations.

Inventory Management

Module	Specifications
Tracked items	Cement bags, stone dust tonnes, sand tonnes, aggregates tonnes, diesel litres, water m3, pallets, moulds, spare parts.
Transactions	Add stock and issue stock to production with supplier, delivery truck number, cost per delivery, production batch link, and notes.
Controls	Low-stock alerts, stock valuation in UGX, daily consumption, days remaining, monthly usage report, reorder level calculation.
Dashboard	Current inventory, stock value, daily consumption value, days of stock remaining, top consumed materials.
Financial connection	Unit costs and stock values support financial cost assumptions and ERP material usage reporting.

Quality Control and Laboratory

Module	Specifications
Batch fields	Batch ID, date, product type, operator, cement used, stone dust used, sand used, water used, mix ratio, curing start date, curing age, test result, approval status.
Tests	Compressive strength, water absorption, density, dimensions, visual defects.
Outputs	Passed batches, failed batches, rejection rate, average strength by product, defect trends, printable test certificate.
Production integration	QC tests are linked to production batches by batch ID and product ID.

ERP

Module	Specifications
Modules	Sales, Inventory, Production, Quality Control, Customers, Suppliers, Expenses, Reports; Dispatch & Delivery integration from the continuation step.
Sales	Customer name, product, quantity, unit price, delivery cost, paid amount, balance.
Production	Daily production, product type, material consumption, machine hours, rejected products.
Reports	Daily sales, monthly profit, inventory usage, production efficiency, customer balances, supplier balances, QC pass rate.
Integration	All modules share local editable state and are structured for PostgreSQL migration.

Market Intelligence

Module	Specifications
Competitors	Competitor ID, name, location, district, region, GPS if available, business type, products, prices, capacity, machine type, delivery radius, contact, links, strengths, weaknesses, threat level, notes, source URL, last updated.

Feature	Description
Price intelligence	Tracks market prices for blocks, pavers, kerbstones, drainage channels, culverts, ready-mix concrete with supplier, location, unit, UGX price, delivery inclusion, date checked, source URL, confidence.
Customer demand	Contractors, hardware stores, property developers, schools, hospitals, churches, NGOs, government projects, fuel stations, hotels, residential builders with likely products, demand, buying power, payment reliability.
Scoring	Opportunity score = demand + margin + strategic value - competition - logistics difficulty.
SWOT	Automatic strengths, weaknesses, opportunities, threats, and response strategy for each competitor.
Dashboards	Competitor count by district, product prices by supplier, average market price, opportunity score, threat distribution, target customer distribution, planned vs competitor price.
Focus areas	Mbarara City, Rwampara, Isingiro, Ntungamo, Bushenyi, Ibanda, Kiruhura, Sheema, Lyantonde, Kabale, Rukungiri.
Seed entries	iConic Concrete, Master Industries, Stone Providers Ltd, Nicomah, BIMS Bloks Mbarara, local block yards, local quarry suppliers, Hima/Tororo/Kampala/Simba Cement distributors, all marked Needs verification.

PDF Master Report

Feature	Description
Required chapters	Cover page, table of contents, executive summary, market analysis, competitor analysis, product portfolio, mix ratios, raw material consumption, machinery, factory layout, utilities, QC, legal roadmap, financial model, risk register, implementation roadmap, East African expansion, appendices.
Format	A4 portrait consultancy report, not slide style; chapters, paragraphs, tables, figures, appendices, page numbers.
Data confidence	Verified / Estimated / Quotation Required / Needs verification labels and assumption notes.
Disclaimer	Market prices and competitor data must be re-verified before investment decisions.
Download	Direct app URL serves the corrected full merged PDF.

PostgreSQL Integration Structure

The platform includes a PostgreSQL-ready schema designed for future ERP integration. Monetary fields are UGX-only, and market intelligence is separated from accounting/ERP tables so unverified records can be governed separately.

products	Product master, portfolio, planned UGX prices, mix ratios, confidence labels.
suppliers	Supplier master for cement, quarry, diesel, water, spares, logistics.
customers	Customer demand, customer type, target products, payment reliability.
production_batches	Batch production, operators, product, material consumption, rejects, curing.
inventory	Stock master, opening/current quantity, unit costs, reorder levels, daily use.
inventory_transactions	Add stock and issue-to-production records linked to suppliers and batches.
sales	Sales records linked to customers and products with UGX price, delivery, paid amount.
expenses	Operating cost records linked to supplier/payee where applicable.
quality_tests	Batch-linked lab tests, strength, absorption, density, dimensions, visual defects, approval.
competitor_records	Market intelligence extension with source tracking and verification status.

Source Files and Download Locations

React platform	outputs/mbarara-concrete-factory-platform/src/App.tsx
Reusable PDF generator	outputs/mbarara-concrete-factory-platform/src/modules/ReportPdfGenerator.tsx
Financial module	outputs/mbarara-concrete-factory-platform/src/modules/FinancialModule.tsx
Inventory module	outputs/mbarara-concrete-factory-platform/src/modules/InventoryModule.tsx
Production module	outputs/mbarara-concrete-factory-platform/src/modules/ProductionModule.tsx
Quality module	outputs/mbarara-concrete-factory-platform/src/modules/QualityModule.tsx
ERP module	outputs/mbarara-concrete-factory-platform/src/modules/ErpModule.tsx
Market intelligence module	outputs/mbarara-concrete-factory-platform/src/modules/MarketIntelligenceModule.tsx
PostgreSQL schema	outputs/mbarara-concrete-factory-platform/src/db/postgres-schema.sql
Corrected browser download	http://127.0.0.1:5177/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf

Corrected Final Download

The browser-served file Mbarara_Integrated_Concrete_Factory_Master_Report.pdf is replaced with the corrected all-prompts merged PDF. It includes this supplement, the recent integrated module master report, and the earlier 194-page investor-grade report.

Direct browser download	http://127.0.0.1:5177/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf
Workspace output	outputs/Mbarara_Integrated_Concrete_Factory_Master_Report.pdf
All-prompts merged copy	outputs/Mbarara_Integrated_Concrete_Factory_ALL_PROMPTS_MERGED_Report.pdf